

OpenArm 2.0 Mechanical Design

Project Scope

This project explores the mechanical housing and exterior integration for the OpenArm 2.0 robotic arm and its mobile manipulator platform. The objective was to develop a manufacturable enclosure system that protects sensors and wiring while maintaining serviceability, structural compatibility, and integration with the existing OpenArm architecture.

The CAD represents a design intent model, with certain features simplified or omitted for clarity and subject to refinement during detailed design and physical validation.

A key focus was not only functional packaging, but also defining a cohesive product direction that moves the system from a raw research platform toward a more complete, demo-ready robotic product.

Design Approach

For this project, I focused on designing a housing system that protects sensors and wiring while remaining practical to manufacture and assemble. The goal was to create a design that feels integrated with the existing OpenArm platform while still allowing access for maintenance and future iteration.

One key lesson from robotics projects is that designs rarely stay identical to their CAD state. Components need to be assembled, serviced, and occasionally replaced, so the housing system was designed with this in mind from the beginning.

The concept sketch defined the overall design language and aesthetic direction. The CAD model then translated this into a manufacturable panel system, with adjustments made to fit the actual arm geometry and estimated joint clearances.

Design Direction (Rationale)

The chosen direction, an organic hexagonal panel language with softened geometry, was selected to balance structural clarity with a more refined product aesthetic. The goal was to move away from an exposed “prototype-looking” robotic arm toward a system that reads as a cohesive, intentional product suitable for demos, research showcases, and public-facing environments.

The hexagonal structure provides a clear geometric system that can be repeated across segments. The matte white and black joint contrast was chosen to emphasize articulation points and create visual hierarchy across moving and static components.

Multiple design directions were explored, with the hexagonal panel system selected for its balance of structure and product clarity. Overall, this direction prioritizes readability, modularity, and visual coherence while still remaining grounded in manufacturable panel-based construction.

Housing Design

The arm housing uses a series of PETG-printed panels that follow the existing geometry of the OpenArm structure. Rather than treating the housing as a separate shell, the intent was to integrate it into the arm while still protecting internal wiring and electronics.

Panels primarily cover exposed regions of the arm and align with existing M6 tap locations where available; not all mounting points are explicitly represented in CAD. In regions without dedicated mounting points, custom 3D printed brackets interface with structural features or existing hardware, avoiding modification to the underlying arm.

Although the CAD is shown as continuous surfaces for visual clarity, production versions would include split lines at service locations. This allows individual sections to be removed independently for maintenance without full disassembly, while keeping structural fasteners visually hidden beneath the exterior skin.

Camera Integration

The system includes two wrist-mounted Arducam cameras near the gripper for close-range manipulation feedback. These are integrated into the arm housing to maintain visibility of grasping tasks while minimizing interference with end-effector motion.

A ceiling-mounted Arducam is attached to a dedicated mini chassis with metal brackets, bolted to the top of the mobile base and angled downward to observe the full workspace. The ceiling Arducam mounting chassis shown in the rendering is a conceptual structural enclosure and is not to scale; in a finalized design, this structure would be significantly more compact and optimized for weight and packaging.

Additionally, a ZED Stereo Camera is mounted on the robot's chassis between the arm mounting regions, positioned at a lower height aligned with the manipulation workspace. This provides stable depth perception of the interaction area without being affected by arm motion.

Camera placement was treated as a tradeoff between visibility, protection, and mechanical packaging, with priority given to stable viewpoints for perception and data collection.

Cable Routing and Service Access

CAN FD and camera wiring route internally through existing channels in the arm structure. The housing panels are shaped to avoid obstructing these routing paths while protecting cables from external contact during operation.

Internal motor and controller geometry was approximated from the provided STEP file. Panels were designed with clearance around joint housings to account for internal components, cable routing, and thermal spacing requirements.

Because physical hardware was not available, routing paths and internal packaging assumptions are based on geometry interpretation rather than full validation. Final verification would require motion testing and physical assembly.

Service access is handled through removable panel sections that allow access to actuators, wiring, and structural fasteners without full disassembly of the housing system.

Manufacturing Considerations

Exterior panels are 3D printed in PETG to balance durability, thermal resistance, and iteration speed. This allows consistent exterior surfacing while keeping production accessible in a small-shop environment.

Sheet metal and machined aluminum are reserved for hidden structural components such as mounting interfaces, brackets, and parts of the mobile base chassis. This separates aesthetic surfacing from load-bearing structure.

Panels mount to existing M6 tap locations where available; additional attachment points are represented using simplified or representative bracket geometries where required, with final placement subject to physical verification. Heat-set inserts will be used in printed parts to improve thread durability.

The design avoids specialized manufacturing requirements, ensuring most components can be produced using standard FDM printing and basic fabrication tools.

Mobile Manipulator Platform

The mobile base uses a compact omnidirectional platform with mecanum wheels, allowing repositioning in tight lab environments without requiring a large turning radius.

A key design influence came from prior FRC experience, where weight distribution and compact packaging were critical for reliable robot performance. This informed the decision to position heavy components such as batteries and compute hardware low in the base to reduce center-of-gravity height and improve stability during arm extension.

The base uses a wide footprint and central mounting interface to distribute load from the arm into the chassis, improving resistance to tipping moments under dynamic operation.

Internally, the base includes dedicated space for batteries, compute hardware, and electronics. This approach was informed by prior FRC experience, particularly in driver station design where compact component packaging and CNC-machined storage features were used to optimize accessibility and weight distribution. The external housing follows the same PETG surface language as the arm, while internal structure would likely be fabricated from sheet metal and machined components for stiffness and load handling.

Assumptions and Limitations

Due to limited access to the physical system, several assumptions were required:

- Internal motor and controller geometry was estimated from STEP data
- Cable routing paths were inferred from visible structure
- Mounting hole alignment requires physical verification
- Camera placements outside the wrist system are conceptual
- Joint clearances were approximated without full motion simulation
- Panel clearances at articulation points would require physical validation

Housing panels are shown in orange in CAD for clarity, distinguishing new components from the original arm geometry. The intended final finish is matte white panels with black joint covers, consistent with the concept direction.

Future Work

If additional time were available, the next steps would be:

- Validate clearances through full motion simulation
- Refine panel split locations based on physical access requirements
- Build and test physical prototypes
- Evaluate thermal performance near actuators and electronics

- Optimize weight and panel thickness for manufacturability
- Validate camera coverage across full workspace

Closing Statement

The objective was to develop a housing and mobile integration system that balances aesthetics, manufacturability, and serviceability. The final design prioritizes realistic fabrication methods while maintaining a cohesive product language across both the arm and mobile platform.